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CC5068NI Cloud Computing and Internet of Things

IoT-Based Health and Room Monitoring System

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I confirm that I understand my coursework needs to be submitted online via MySecondTeacher under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a marks of zero will be awarded.

Acknowledgement

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Abstract

The advancement and integration of Internet of Things technologies in remote healthcare have significantly enhanced the way the healthcare system operates. However, the technical aspects of it still consist of critical gaps and the project aims to address those limitations. This project focuses on building an affordable and lightweight IoT system that monitors the patients' health and the room environment at the same time.

The system uses an Arduino Uno connected to a DHT11 sensor for the humidity and temperature and a pulse sensor for tracking the heart rate. The system processes sensor data in real-time and filters out errors while integrating an automatic alarm if the readings hit dangerous levels. It gives immediate health updates and emergency alerts using an LCD screen.

This system is used to help patients who require extensive care such as the elderly, disabled or non-verbal patients to communicate their needs and provide more secure and effective experience.

Table of Contents

1	Introduction	1
1.1	Project Overview	2
1.2	Current Scenario	3
1.3	Problem Statement	4
1.4	Aim and Objectives	4
1.5	Terms and Terminologies	5
2	Background and System Design	6
2.1	System Overview	6
2.2	System Design and Architectural Diagrams	7
2.2.1	System Architecture	7
2.2.2	Block Diagram	8
2.3	Detailed Requirement analysis	9
3	Development Process	10
3.1	Step 1: Component Gathering	10
3.1.1	Hardware Arrangement.....	10
3.2	Step 2: Wiring and Assembling Hardware.....	10
3.2.1	Power Supply Wiring.....	10
3.2.2	Hardware and Arduino Assembling	11
3.3	Step 3: Software Setup	12
3.3.1	Initialize Arduino IDE.....	12
3.3.2	Library Installation.....	12
3.4	Step 4: Writing and Uploading code.....	13
3.4.1	Code	13
3.4.2	Uploading the code.....	13
3.5	Step 5: Testing and verifying the system.....	13
3.5.1	Data from Sensors	13
3.5.2	Prototype testing	13
3.5.3	Improvements	13
3.5.4	Flowchart	14

3.5.5	Circuit Diagram	15
3.6	Hardware Components	19
3.6.1	Arduino UNO	19
3.6.2	I2C LCD Adapter.....	20
3.6.3	DHT11 Sensor	21
3.6.4	Pulse Sensor	22
3.6.5	Jumper Wires.....	22
3.6.6	Breadboard	23
3.6.7	Arduino USB Cable.....	24
3.6.8	Liquid-Crystal Display	25
3.7	Software and Tools	26
3.7.1	Arduino IDE	26
4	Results and Findings.....	28
4.1	Testing 1: System Testing	28
4.2	Testing 2: Software testing	31
4.3	Unit Testing	32
4.3.1	Testing 1: Humidity testing with lighter	32
4.3.2	Testing 2: Temperature testing with lighter	34
4.3.3	Testing 3: LCD Display testing	36
4.3.4	Testing 4: Pulse Sensor testing.....	37
4.4	Failed Test Cases	38
4.4.1	Pulse Sensor Failure Test Cases	38
4.4.2	DHT Sensor Failure Test Case	39
5	Further Work	40
6	Conclusion	41
7	References.....	42
8	Appendices	44
8.1	Appendix A.....	44
8.2	Arduino UNO.....	44
8.3	Pulse Sensor.....	44

8.4	Liquid Crystal Display	44
8.5	Appendix B.....	45
8.6	DHT11 SETUP	45
8.7	PULSE SENSOR SETUP	45
8.8	LCD SETUP	45
8.9	VARIABLES	46
8.10	SETUP.....	46
8.11	LOOP	48

Table of Figures

Figure 1: IoT for Healthcare (Singh, 2020)	1
Figure 2: System Design & Architectural Diagrams.....	7
Figure 3: Block Diagram.....	8
Figure 4: Flow Chart.....	14
Figure 5: Circuit diagram of overall system	15
Figure 6: Circuit Diagram of Arduino and Breadboard.....	16
Figure 7: Circuit Diagram of DHT11 with Arduino and Breadboard	17
Figure 8: Circuit Diagram of Pulse sensor with Arduino and Breadboard.....	17
Figure 9: Circuit Diagram of LCD I2C adapter with Arduino and Breadboard.....	18
Figure 10: Arduino UNO	19
Figure 11: LCD I2C Module IIC/I2C Serial Interface Adapter	20
Figure 12:DHT11 Sensor.....	21
Figure 13: Pulse Sensor.....	22
Figure 14: Male and Female Jumper Wires	22
Figure 15: Structure and Connectivity of a Breadboard.....	23
Figure 16: Arduino Uno with USB Connection.....	24
Figure 17: LCD Screen Connection	25
Figure 18: Arduino IDE	27
Figure 19: System Testing (1).....	29
Figure 20: System Testing (2).....	30
Figure 21: Humidity Reading before over exposure to lighter	32
Figure 22: Humidity reading after over exposure to lighter.....	33
Figure 23: Temperature before over exposure to lighter.....	34
Figure 24: Temperature after over exposure of lighter.....	35
Figure 25: LCD display testing	36
Figure 26: Pulse Sensor Testing.....	37
Figure 27: Pulse sensor VCC wire unplugged.....	38
Figure 28: DHT11 VCC wire unplugged	39

Table of Tables

Table 1: DHT11 connection to Arduino	11
Table 2: Wiring between Pulse sensor and Arduino	11
Table 3: Wiring between I2C LCD adapter and Arduino	11
Table 4: Test 1 System Testing	28
Table 5: Test 2 Software Code Test	31
Table 6: Test 3 Humidity Testing	32
Table 7: Test 5 LCD display testing	36
Table 8 Pulse Sensor Failure Test Table	38
Table 9 DHT Sensor Failure Test Table	39

1 Introduction

The rapid development of IOT has enabled the advancement of Health Monitoring systems in the world. With the increasing prevalence of diseases and the need for real-time health monitoring, tracking and preventive healthcare, technology and development play a crucial role in modern health solutions. Alongside the health monitoring, the necessity of one to alert staff or helpers to their room conditions in hospital or home settings has also simultaneously grown.



Figure 1: IoT for Healthcare (Singh, 2020)

The modern period of the health industry needs a low-cost and lightweight system to track the health parameters of a patient and simultaneously record the environmental conditions of the room they are in. The creation of an IOT-based health and room monitoring system will help to bridge communication between the patient and the healthcare staff. The reduction of unnecessary hospital visits and proactive action by the helpers are some of the few advantages of such a system (Al-Wesabi, et al., 2025) (V & D, 2022).

1.1 Project Overview

Individuals who face physical or mental disabilities are highly at risk of undetected health deterioration in unsuitable room conditions. Such patients with limited mobility cannot seek immediate help to control the AC or communicate their uneasiness. This will lead the patient to go through serious medical problems.

By the integration of IOT technologies like sensors, processors, and actuators, this project introduces an IOT-based Health and Room monitoring system. Its purpose is to develop a System that continuously records a user's physical condition and provides timely feedback, and alerts. Additionally, a room monitoring system will show the room condition of the patient in real-time and alert the health authorities when the temperature is too high or too low.

It's dual functionality of recording patient's health parameters like heart rate and room temperatures will alert the health staff to prompt action. This way, patients with any type of problem will recover easily and stay comfortably (V & D, 2022) (Al-Wesabi, et al., 2025).

1.2 Current Scenario

The health and room monitoring systems in the current time have been more real-time, cloud-connected and AI-driven. The health monitoring systems have been implemented on wearable devices and ambient sensors that record the patient's health parameters. The data from these devices are then sent to the health staff or caregivers for analyzation. This type of device does real-time monitoring and gives timely alerts for those who have difficulty in going on regular hospital visits.

The room monitoring systems are used to track the air fluctuating temperature of the rooms. They are focused on the conservation of energy so that the battery or low-power chips do not finish rapidly. There is also implementation of AI models in these new systems where they can predict when the room's air quality will get worse.

Despite the benefits, there are limitations that lessen the trust and reliability of these systems. The data accuracy cannot be reliable as the readings in the sensors may show temperatures of the room higher by 1 or 2 degrees than the original. The IOT devices must be light-weight and low-cost, so there is not enough implementation of the security features. There must be stronger encryption for the protection of sensitive health and environmental data.

Therefore, the current scenario of the health and room monitoring system has been helpful to many even if it has some major limitations that question their reliability and accuracy.

(Witczak & Szymoniak, 2024)

1.3 Problem Statement

Undetected health deterioration and unsuitable environmental conditions are the two major risks faced by patients in old age homes, hospitals, and home-based care settings. The traditional monitoring systems only focused on the vital signs, meanwhile the room's environmental condition was dealt with manually. Small yet critical signs like irregular heart rate, added with mental discomfort due to uneasy environmental conditions were not considered. This is a struggle for the elderly and the disabled people, who are unable to properly communicate with the health staff about their discomforts and needs. This alerting problem requires an IoT device that will record the patient's health parameters and simultaneously alert the health authorities of irregular room conditions (V & D, 2022) (Al-Wesabi, et al., 2025).

1.4 Aim and Objectives

Aim

The aim of this project is to design and develop an IOT-based healthcare system that includes both hardware and software applications which allows real-time tracking of room conditions, patient vital signs and providing alerts according to it while understanding the core concepts of IOT and their implementations and solving real-world problems.

Objectives

The objectives of this project are listed below:

- Developing a health Monitoring System that observes basic health parameters such as heart rate, temperature rise or fall, and activity level.
- Providing real-time data for individuals using IoT technology.
- Implementing an alert when any abnormal health conditions are spotted (Heart rate).
- Storing and managing records of health data for future inquiries.
- Gaining familiarity with the sensors, programming and IoT architecture applied in the real-world to solve problems.

1.5 Terms and Terminologies

The following are the terms and terminologies that have been used in the report:

Terms	Description
Sensor	Sensors are those devices that take external input and respond with an electric signal. The external input can be light, heat, touch, movement or pressure for the sensor to react. Some of the examples are Chemical sensor, gas sensors and pulse sensors (Subramanian & David , 2021).
Microprocessor	Microprocessors are single-chip computing devices that connect the hardware and the software to conduct specific tasks efficiently. This has the integration of three components, they are CPU, RAM, ROM and I/O interfaces. Some examples are Arduino, ESP32, STM32 and more (Vel, 2025).
Circuit	A circuit is defined as a circular path that has interconnected electrical components like capacitors, resistors and inductors from which electric current is passes through (Tooley, 2023).

2 Background and System Design

2.1 System Overview

The health and room monitoring system includes several IOT based technologies to efficiently monitor patients and collect data. The system aims to improve healthcare delivery, and the care provided to customers. It provides timely alerts, real-time feedback, and data visualization. The system ensures that those patients who require critical care are monitored properly for any health complication signs. This helps in reducing the response time and potentially improve patient outcomes.

As room temperature also plays a critical role in maintaining the comfort of the patient and indirectly influencing the health of the patient, this system helps to ensure that if any irregularities occur in the room temperature the responsible authorities will be notified. This system helps to improve the patient's care and ensures that both physical and environmental well-being are actively monitored and managed.

In addition, the system also integrates with IOT technologies to enable continuous monitoring and analysis of data related to the patients' health, which supports the early detection of potential health issues before they escalate. The data collected from the sensors can help forecast changes in the patient's condition based on factors such as heart rate, body temperature, or environmental factors. The combination of health and environmental monitoring systems aims to enhance both the quality and efficiency of patient care and contribute to the healthcare industry.

2.2 System Design and Architectural Diagrams

2.2.1 System Architecture

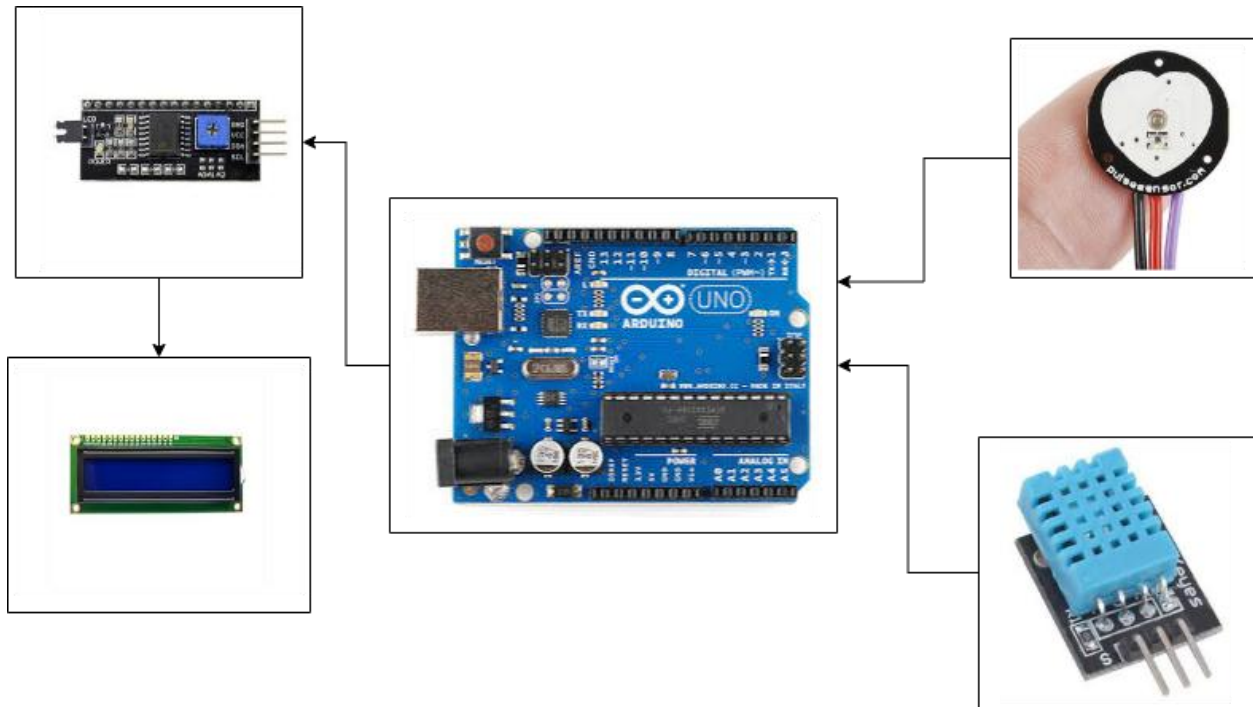


Figure 2: System Design & Architectural Diagrams

2.2.2 Block Diagram

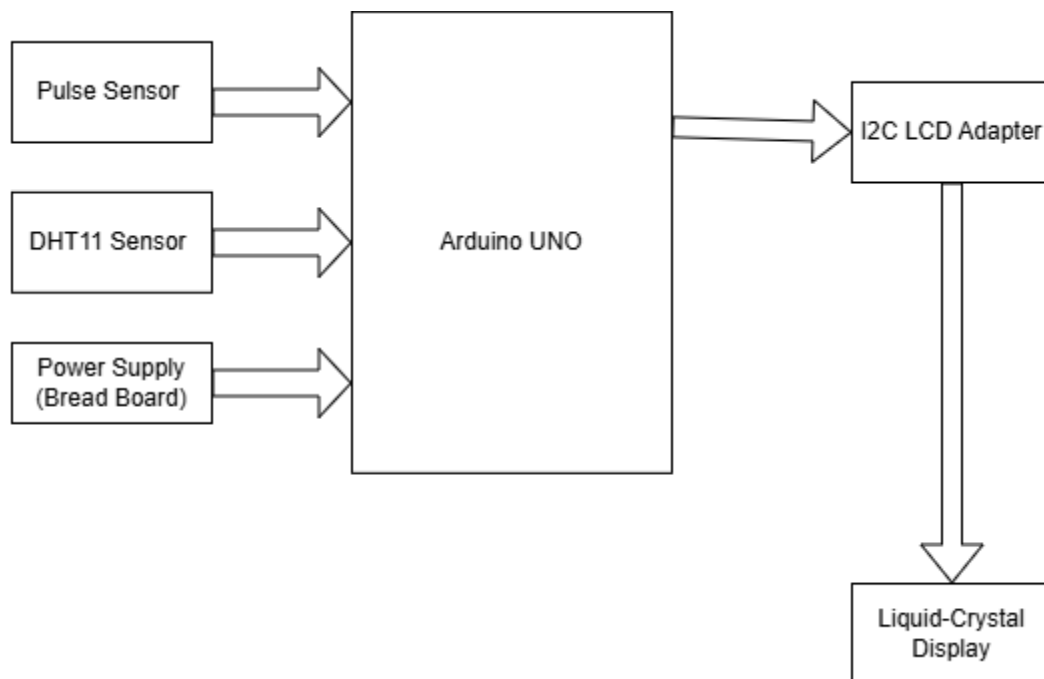


Figure 3: Block Diagram

2.3 Detailed Requirement analysis

The essential things required in the project are explained below:

1. **Health Monitoring:** The system is required to track the real-time heart rate using a pulse monitor and measuring the body temperature.
2. **Room Monitoring:** The system is required to track the room temperature and humidity using DHT11. The sensors are required to detect temperature fluctuations.
3. **Data Collection:** The data from the sensors should be collected using an Arduino UNO microcontroller. The data can later be stored securely and used for further diagnosis of the patients.
4. **Alert System:** If something is detected, there should be immediate alerts shown in the LCD screen, such as in the case of abnormal heart rate or increase in humidity of the room.
5. **Power Management:** The system should include proper power management so that even in the case of emergency the system can still operate, and the data is not lost as well.

3 Development Process

3.1 Step 1: Component Gathering

3.1.1 Hardware Arrangement

The first step is to gather the necessary hardware components required for the project. These include the following components for the project:

- Pulse Sensor for heart rate measurement
- DHT11 for the humidity and room temperature
- Arduino UNO for the microcontroller
- Breadboard for the power supply
- I2C LCD adapter for data display
- Wires for connection
- Arduino USB cable

3.2 Step 2: Wiring and Assembling Hardware

3.2.1 Power Supply Wiring

The positive power rail of the bread board must be connected to 5V pin of Arduino's power header side. And the negative power rail of the bread must be connected to GND pin of the power header side of Arduino. The purpose of this connection is for multiple sensors and actuators to connect to 5V and GND in Arduino through breadboard power rails simultaneously.

3.2.2 Hardware and Arduino Assembling

The following tables show the connections between the hardware and Arduino:

Table 1: DHT11 connection to Arduino

Components	Pins	Connect to Arduino pins
DHT11	VCC	5V (Power header)
	GND	GND (Power header)
	DATA	D2 (Digital Header)

Table 2: Wiring between Pulse sensor and Arduino

Components	Pins	Connect to Arduino pins
Pulse Sensor	VCC	5V (Positive rail)
	GND	GND (Negative rail)
	SIGNAL	A1 (Power header)

Table 3: Wiring between I2C LCD adapter and Arduino

Components	Pins	Connect to Arduino pins
I2C LCD Adapter	VCC	5V (Positive rail)
	GND	GND (Negative rail)
	SDA	A4 (Power header)
	SCL	A5 (Power header)

3.3 Step 3: Software Setup

3.3.1 Initialize Arduino IDE

Arduino IDE must be initialized to run the codes for system functioning. After the software has been opened, the user must click the “Tools” icon and select the board “Arduino UNO”.

3.3.2 Library Installation

To install the libraries, the “Sketch” button on the top should be clicked which will give the option to “manage library”. By clicking on that option, a search bar will appear where libraries can be searched to install. The following are the libraries that must be installed:

Libraries	Purpose
LiquidCrystal_I2C.h	This library helps to control the LCD display.
DHT11	This helps to control DHT11 sensor
PulseSensorPlayground.h	It is used for controlling pulse sensors.

There is also a built-in library called “Wire.h” that is used for I2C communication for LCD backend tasks.

3.4 Step 4: Writing and Uploading code

3.4.1 Code

This code is utilized to run the health and room monitoring system:

[\(See Appendix for code\)](#)

3.4.2 Uploading the code

First, the Arduino USB cable is connected between the laptop and the Arduino. This allows the code to be directed to the controller. After the code is uploaded, the Arduino will activate by signaling a red light.

3.5 Step 5: Testing and verifying the system

3.5.1 Data from Sensors

The system is designed to continuously collect data from the sensors in real-time. As the data collected is very crucial for the treatment of patients, the data is collected continuously. The Pulse Sensor collects heart rate data from a finger touch and provides readings of BPM. DHT11 is used to track the room temperature in Celsius and humidity in percentage of the room.

3.5.2 Prototype testing

After the system is compiled and programmed, testing is conducted to ensure that it works properly.

3.5.3 Improvements

If there are any irregularities found during the testing or if the system is not working properly, it's identified and fixed.

3.5.4 Flowchart

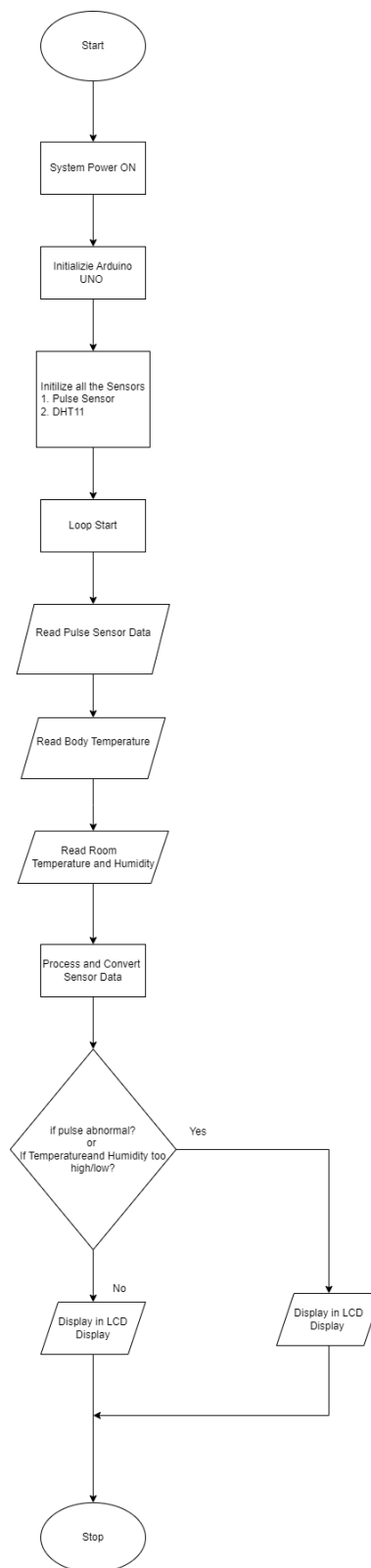


Figure 4: Flow Chart

3.5.5 Circuit Diagram

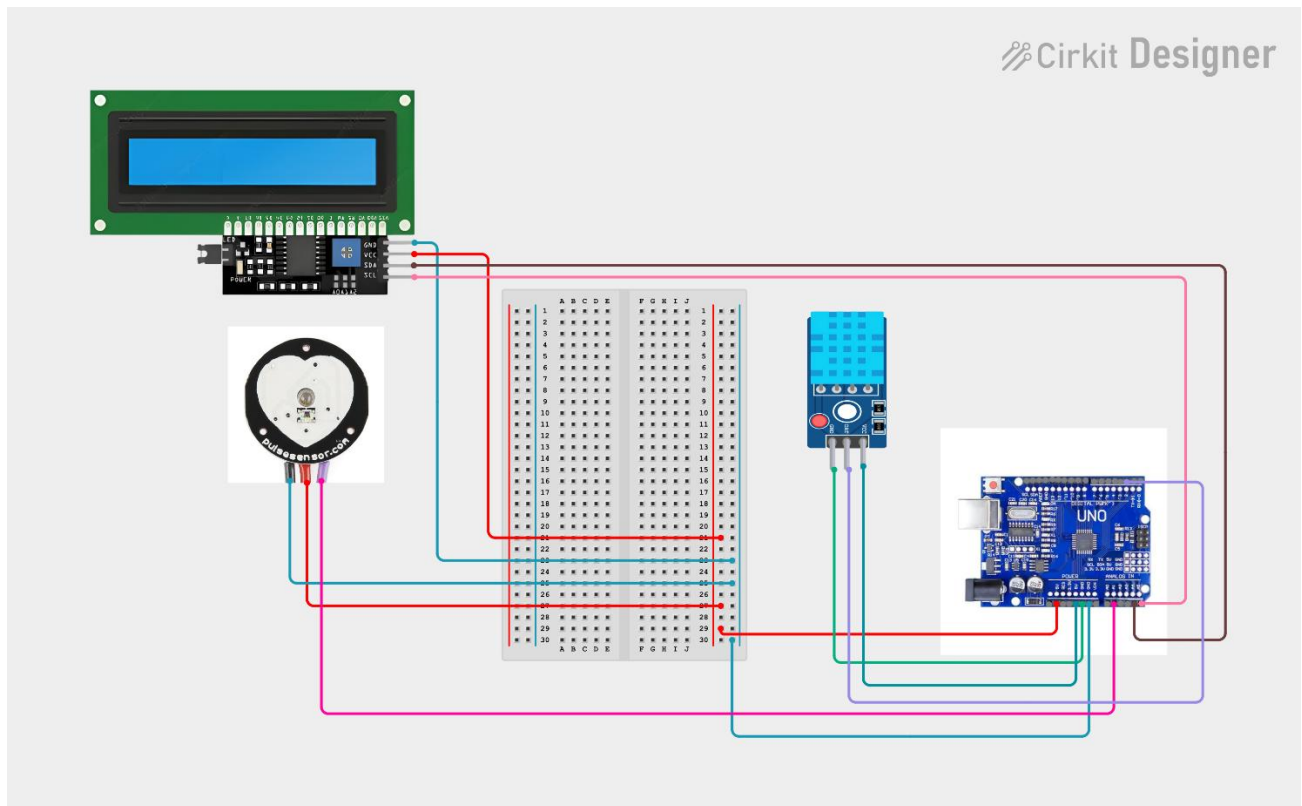


Figure 5: Circuit diagram of overall system

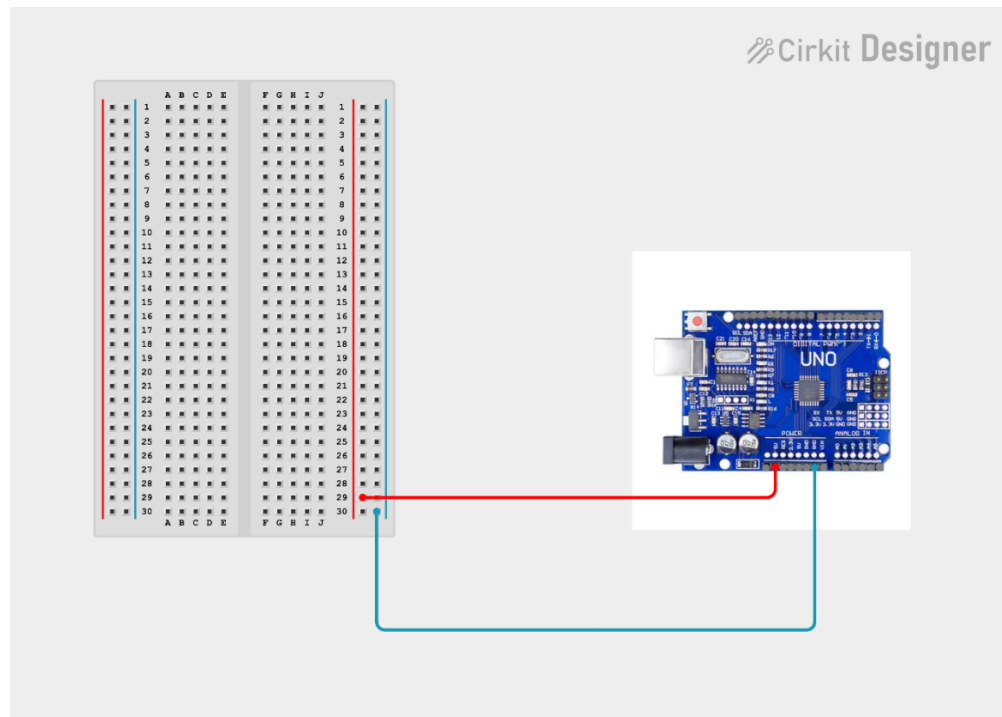


Figure 6: Circuit Diagram of Arduino and Breadboard

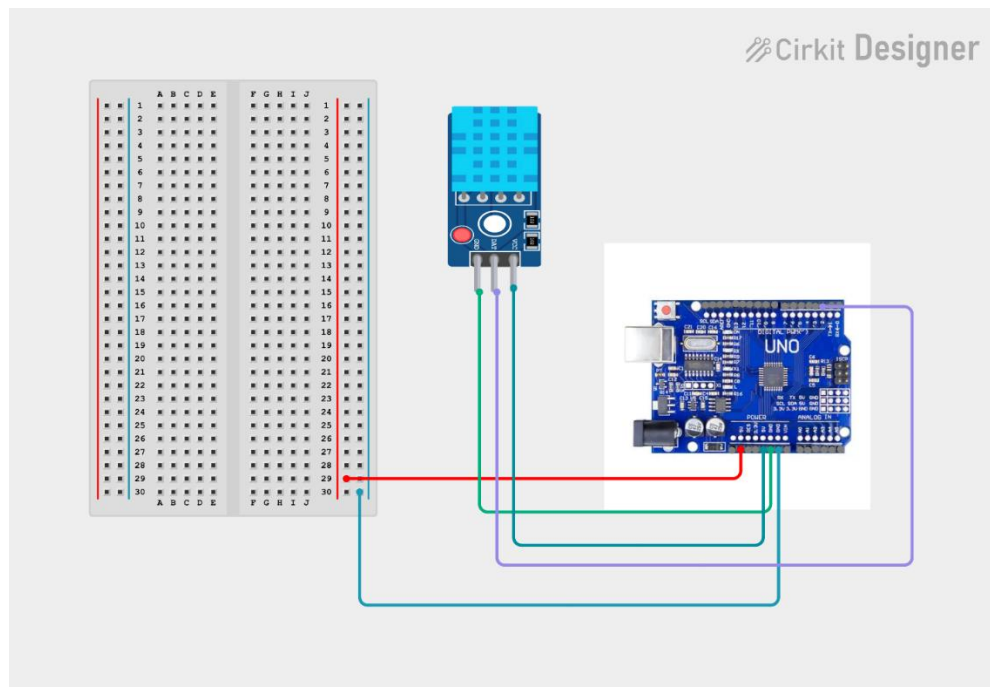


Figure 7: Circuit Diagram of DHT11 with Arduino and Breadboard

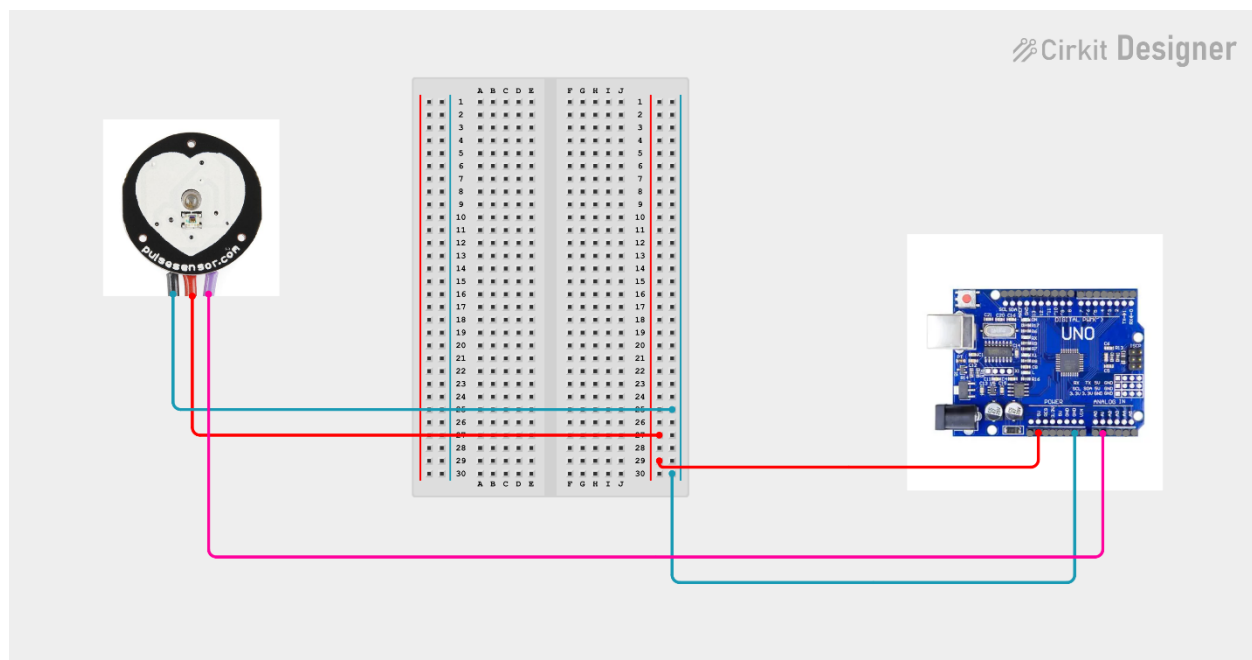


Figure 8: Circuit Diagram of Pulse sensor with Arduino and Breadboard

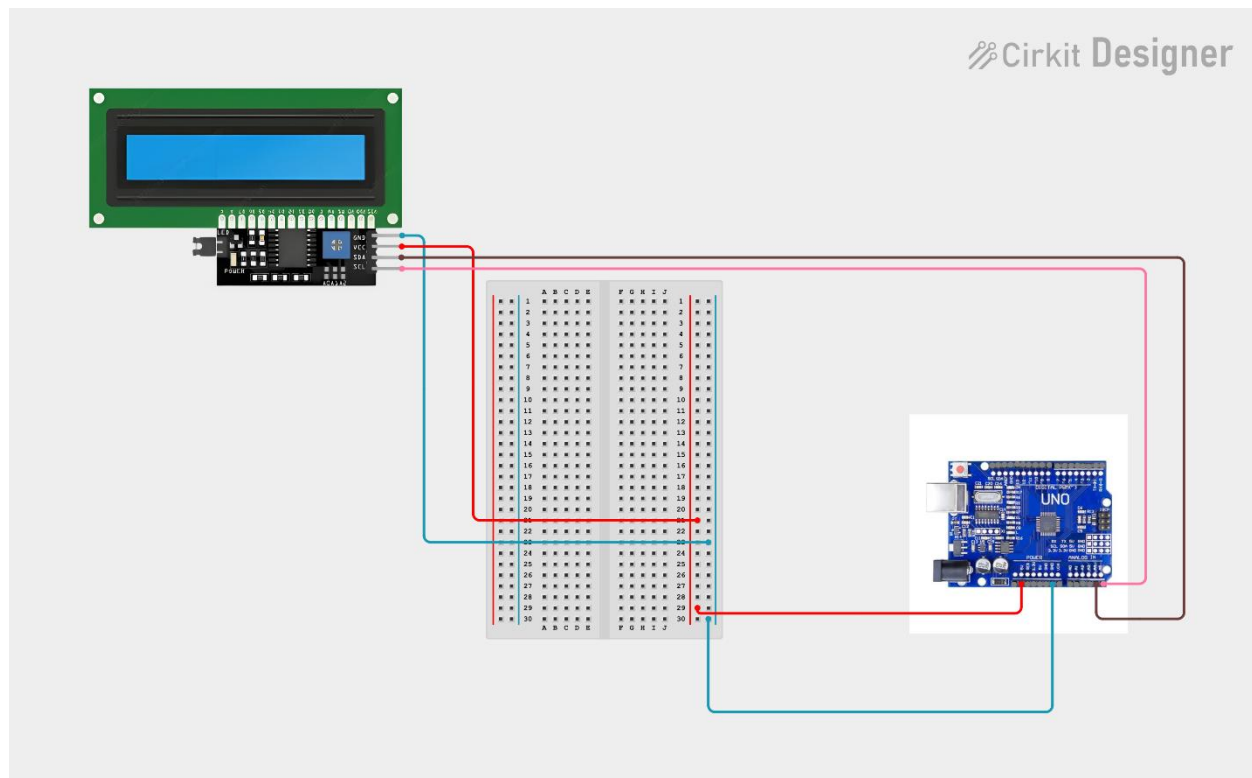


Figure 9: Circuit Diagram of LCD I2C adapter with Arduino and Breadboard

3.6 Hardware Components

The hardware components represent the physical part of the Health and Room Monitoring System. These components are responsible for sensing, processing, and displaying real-time data from the patient's body and the surrounding environment.

3.6.1 Arduino UNO

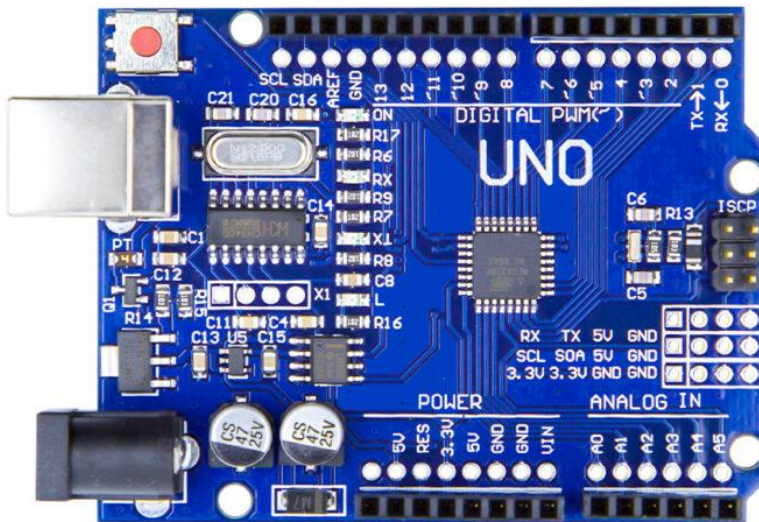


Figure 10: Arduino UNO

The Arduino UNO is the main microcontroller used in the system. It acts as the brain of the entire setup by receiving data from sensors, processes it, and sends it to the output devices (Arduino, 2023).

For further information it is in [Appendix A](#)

3.6.2 I2C LCD Adapter

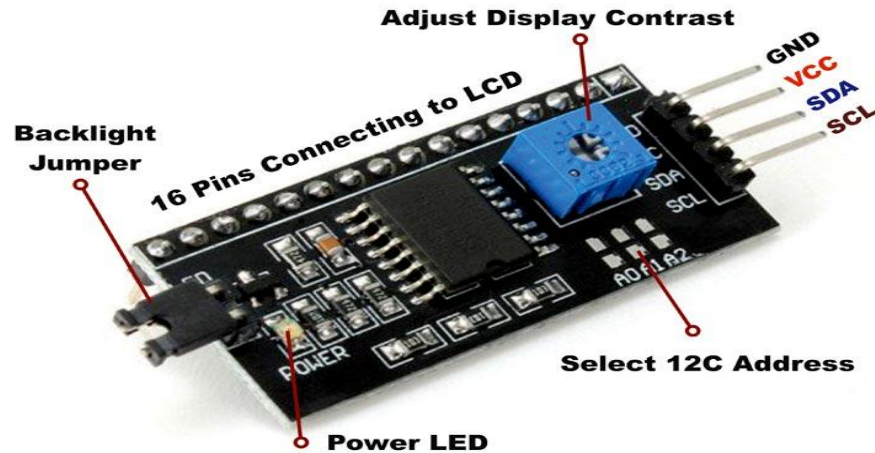


Figure 11: LCD I2C Module IIC/I2C Serial Interface Adapter

The I2C stands for “Inter-Integrated Circuit”, this LCD Adapter is used to make the process of connecting the Arduino UNO and the LCD simpler. In general, an LCD needs many pins however, using the I2C adapter allows us to connect only two pins for communication, which are Serial Data (SDA) and Serial Clock (SCL). The I2C adapter will help our system display real-time information of temperature, humidity, and heart rate readings (Swagatam, 2025).

3.6.3 DHT11 Sensor

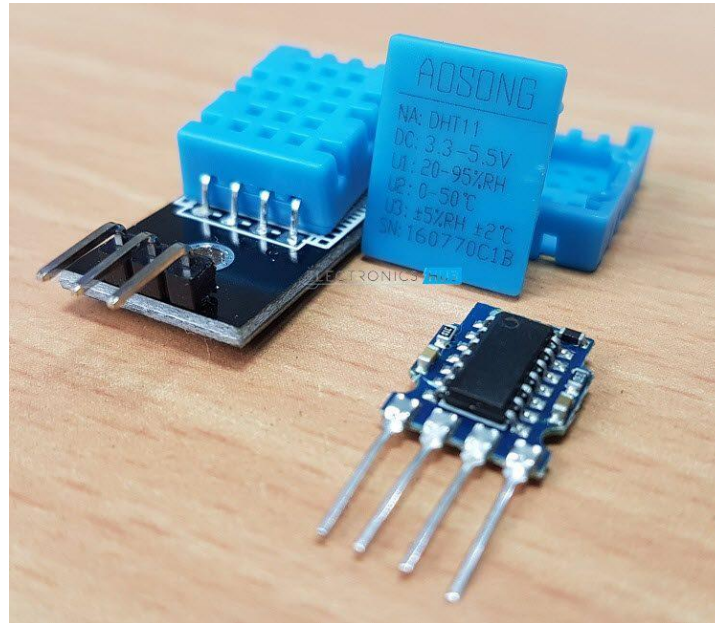


Figure 12:DHT11 Sensor

DHT11 is a low-cost sensor used to measure temperature and humidity. It is widely used in Arduino, Raspberry Pi, and IoT projects because of its simple design and low power consumption (ElectronicsHub, 2024).

3.6.4 Pulse Sensor

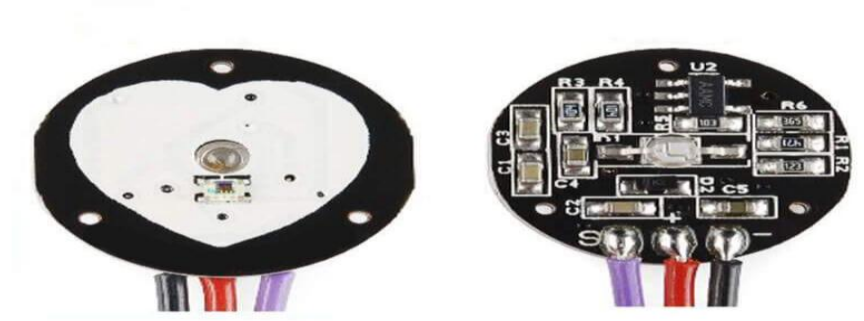


Figure 13: Pulse Sensor

A pulse sensor, also known as a heartbeat or heart rate sensor that reflects to detect the pulse from a fingertip. This sensor is responsible for measuring the heart rate of the patients (Measure Heart Rate, 2026).

Further information about pulse Sensor is in [Appendix A](#)

3.6.5 Jumper Wires

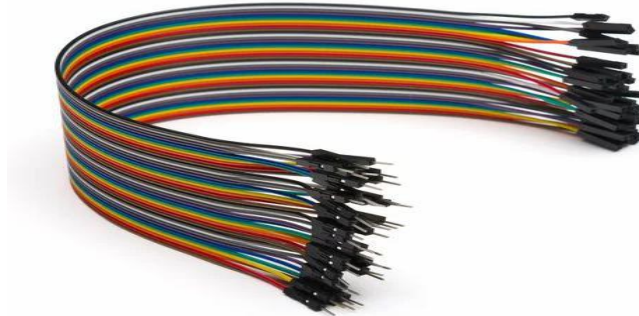


Figure 14: Male and Female Jumper Wires

Jumper wires are tiny metal connectors that are used to close or open a circuit part. These Jumper wires enable the connection between different components of the breadboard and Arduino UNO. They allow easy connections that can easily be changed or re-arranged to prototype the system (Wiltronics, 2022).

3.6.6 Breadboard

The breadboard is used for assembling the circuit without permanent soldering. It allows easy placement and connection of electronic components using Jumper wires.

The breadboard helps to organize the connections between Arduino, sensors, and other devices (Ohmify, 2023).

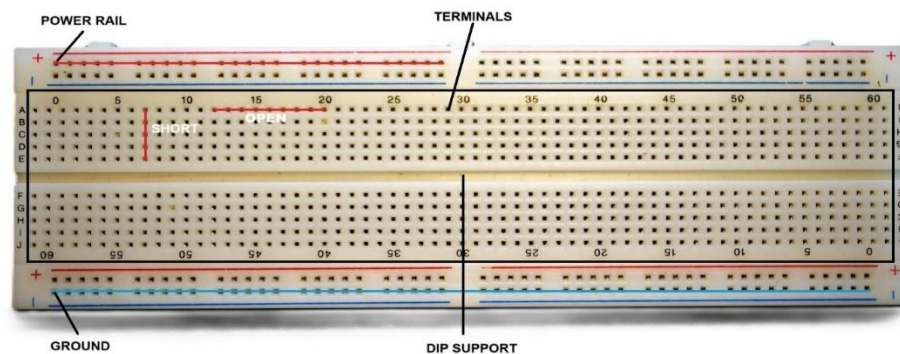


Figure 15: Structure and Connectivity of a Breadboard

3.6.7 Arduino USB Cable

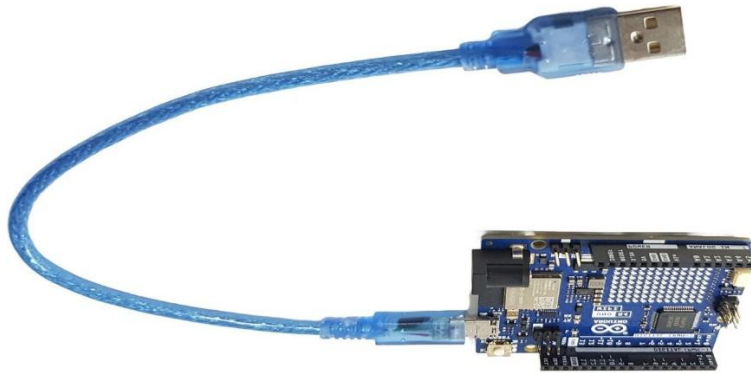


Figure 16: Arduino Uno with USB Connection

The Arduino USB Cable is the one that connects the Arduino UNO to the computer. The functions of the Arduino USB cable include transferring the program(code) to the Arduino board and powering it. Once the system is programmed, the Arduino can also be powered using other sources if needed (Arduino, 2023).

3.6.8 Liquid-Crystal Display



Figure 17: LCD Screen Connection

The Liquid Crystal Display (LCD) is a flat-panel display technology that utilizes the light-altering capabilities of liquid crystals along with backlight to create images. They are also energy efficient, thus becoming increasingly popular in use (Encyclopaedia Britannica , 2024).

Furthermore, information is in [Appendix A](#)

3.7 Software and Tools

The software and tools play a crucial role in the Health and Room Monitoring System by enabling data processing, visualization, storage, and communication between hardware components and users. These tools ensure that the collected data is efficiently managed and presented in a user-friendly manner.

3.7.1 Arduino IDE

Arduino IDE (Integrated Development Environment) is the official development tool used in coding, compiling, and uploading codes into Arduino microcontroller boards like Arduino UNO. The platform offers a simple user interface for programming in C and C++ languages. The software comes with necessary tools including the code editor, compiler, and serial monitor which are used for programming, debugging, and real-time data observation.

In this project, Arduino IDE is utilized to develop programs that will enable DHT11 and pulse sensor readings, process them in the Arduino UNO microcontroller, and control output devices like the LCD. In addition, uploading program code becomes convenient since the IDE enables communication with the microcontroller through the USB interface (Arduino IDE, 2023).

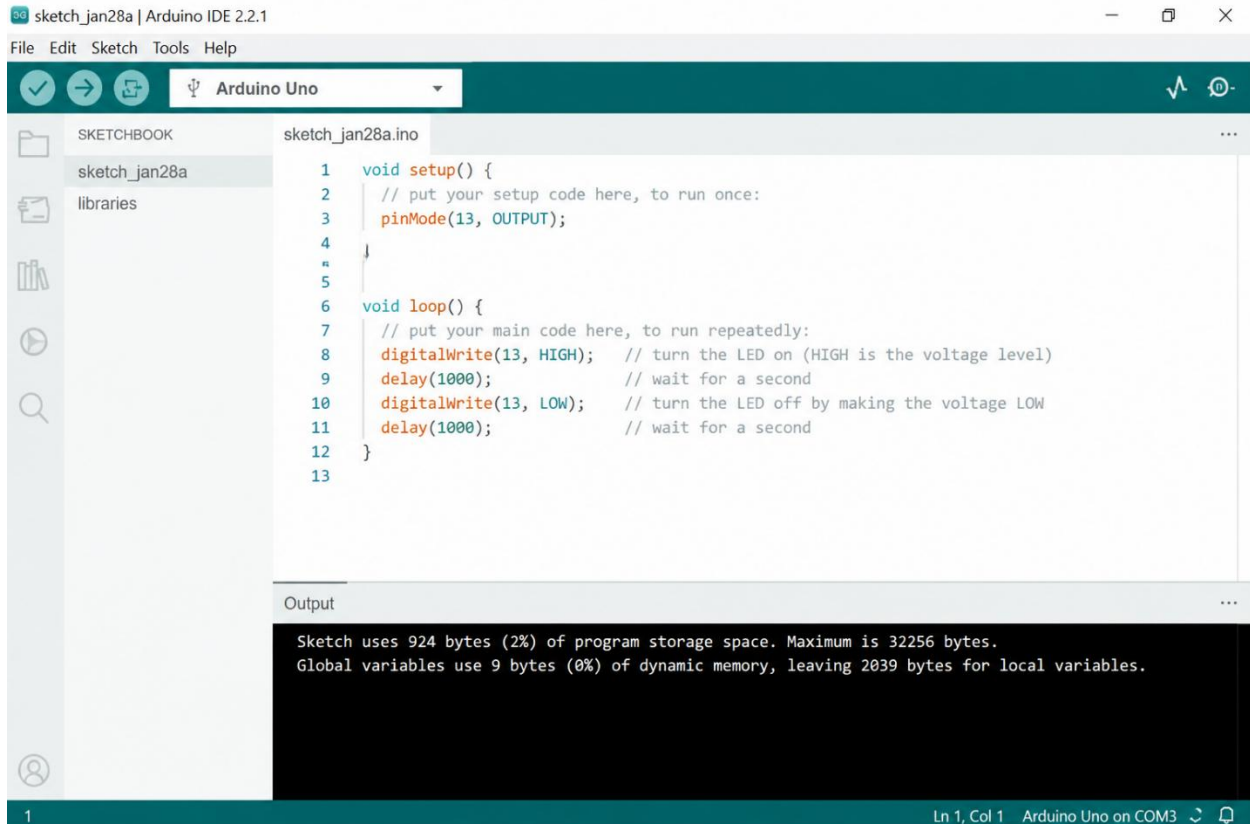


Figure 18: Arduino IDE

4 Results and Findings

4.1 Testing 1: System Testing

Table 4: Test 1 System Testing

Components	Test Objective	Test Description	Expected outcome	Result
DHT11	To check humidity and Room temperature	DHT11 is used to detect real time information and data for temperature and humidity of the room.	DHT11 detected Humidity and Temperature of the room.	Passed.
Pulse Sensor	To check Heart beats (Pulse)	The pulse sensor is used to detect the heartbeat of a human body, by using fingerprint.	Pulse Sensor detected the heartrate of the user using fingerprints accurately.	Passed.
Liquid Crystal Display (LCD)	To display the data and information passed through I2C Adapter	Displays the data and information passed through I2C Adapter.	Displayed the data of temperature, humidity and Pulse.	Passed.

The evidence of the system testing is below:

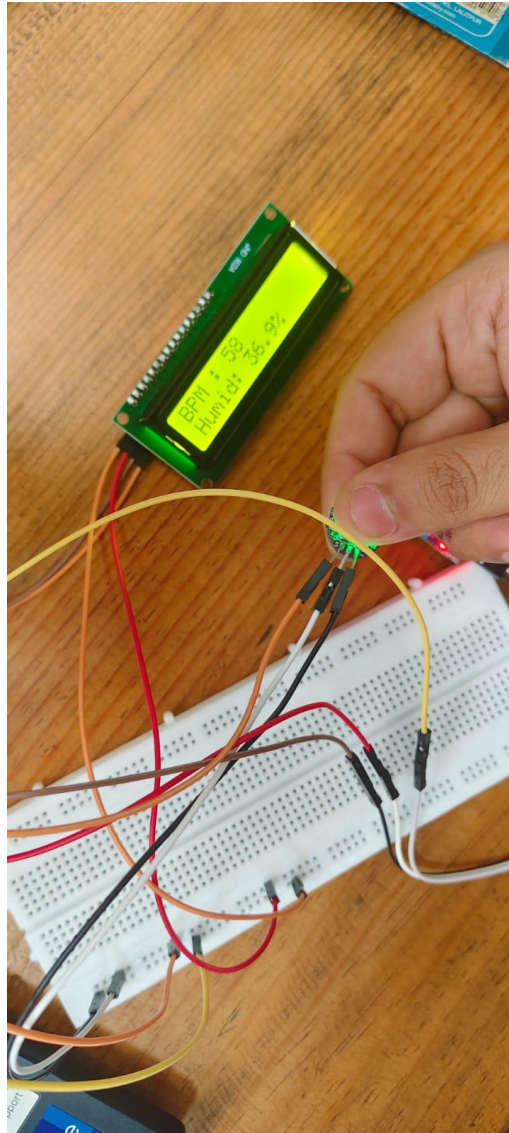


Figure 19: System Testing (1)

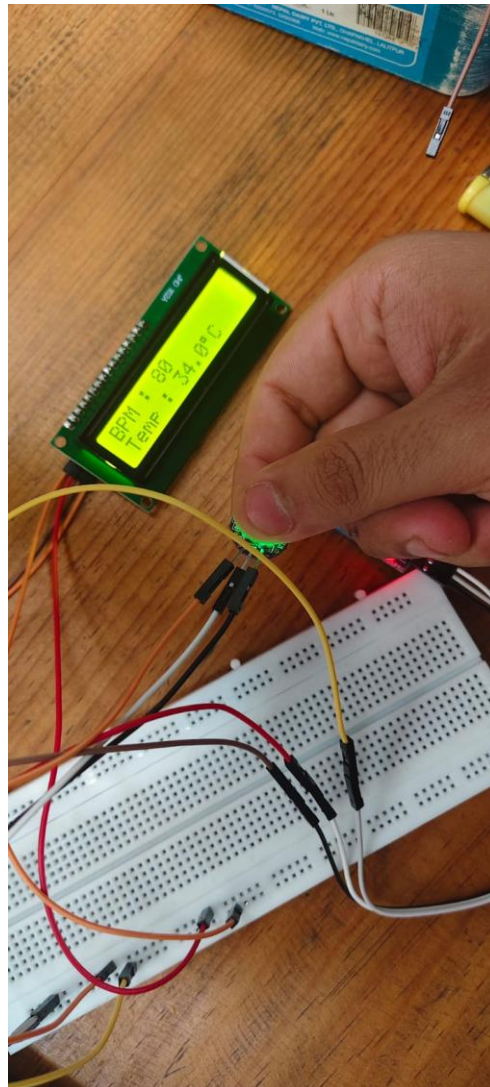


Figure 20: System Testing (2)

4.2 Testing 2: Software testing

Table 5: Test 2 Software Code Test

Components	Test Objective	Expected Outcome	Result
Arduino IDE	To test the code for the system (Health and Room Monitoring)	The code should run and provide outcome in hardware.	Passed.

The code is in [Appendix A](#)

4.3 Unit Testing

4.3.1 Testing 1: Humidity testing with lighter

Table 6: Test 3 Humidity Testing

Component	Test Objective	Expected Outcome	Actual Outcome	Result
DHT11	To test the humidity level by using lighter	The LCD should show how humidity reading becomes less as the exposure to the lighter increases.	The LCD showed how humidity reading became less as the exposure to the lighter increases.	Passed

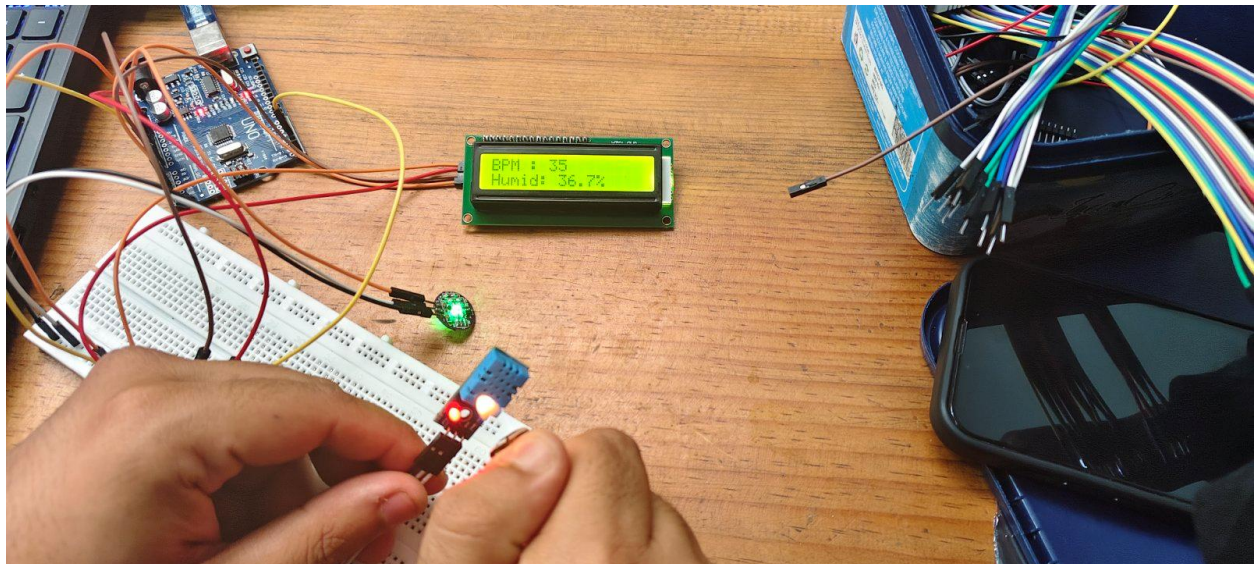


Figure 21: Humidity Reading before over exposure to lighter

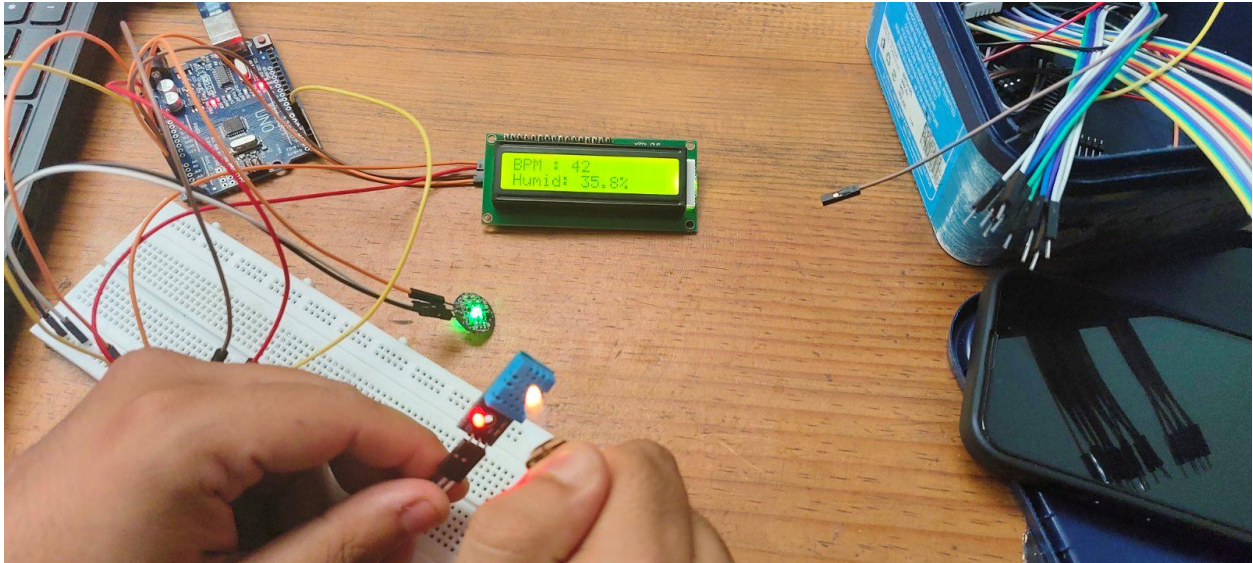


Figure 22: Humidity reading after over exposure to lighter

4.3.2 Testing 2: Temperature testing with lighter

Component	Test Objective	Expected Outcome	Actual Outcome	Result
DHT11	To test the temperature level by using lighter	The LCD should show how the temperature becomes more as the light from the lighter is sensed by DHT11.	The LCD screen showed the correct readings of the temperature when lighter was closer.	Passed

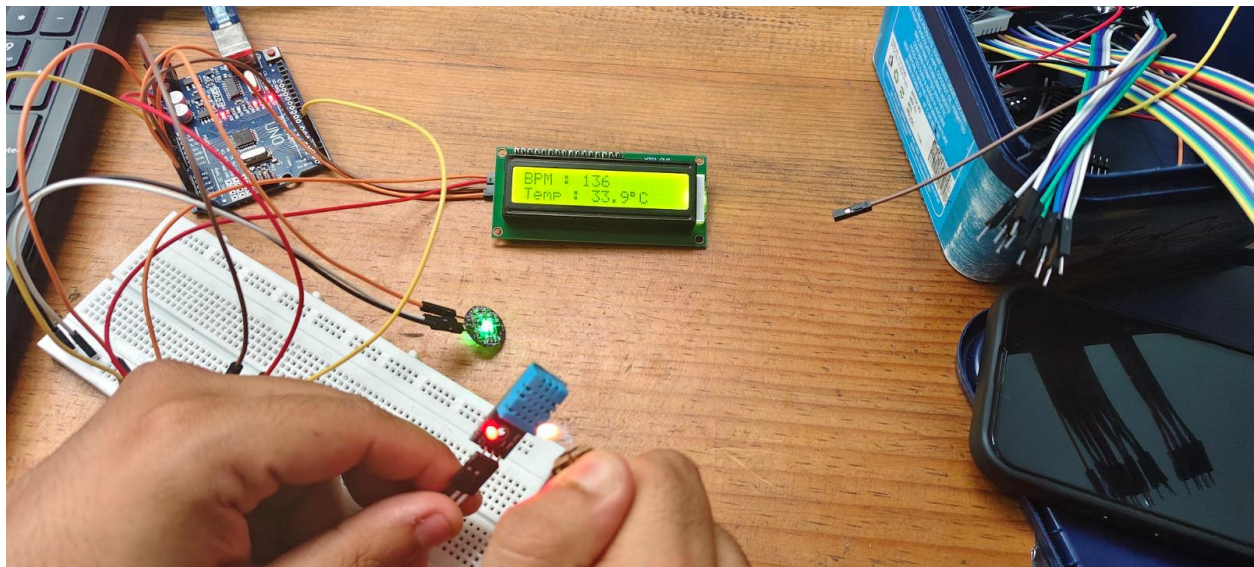


Figure 23: Temperature before over exposure to lighter

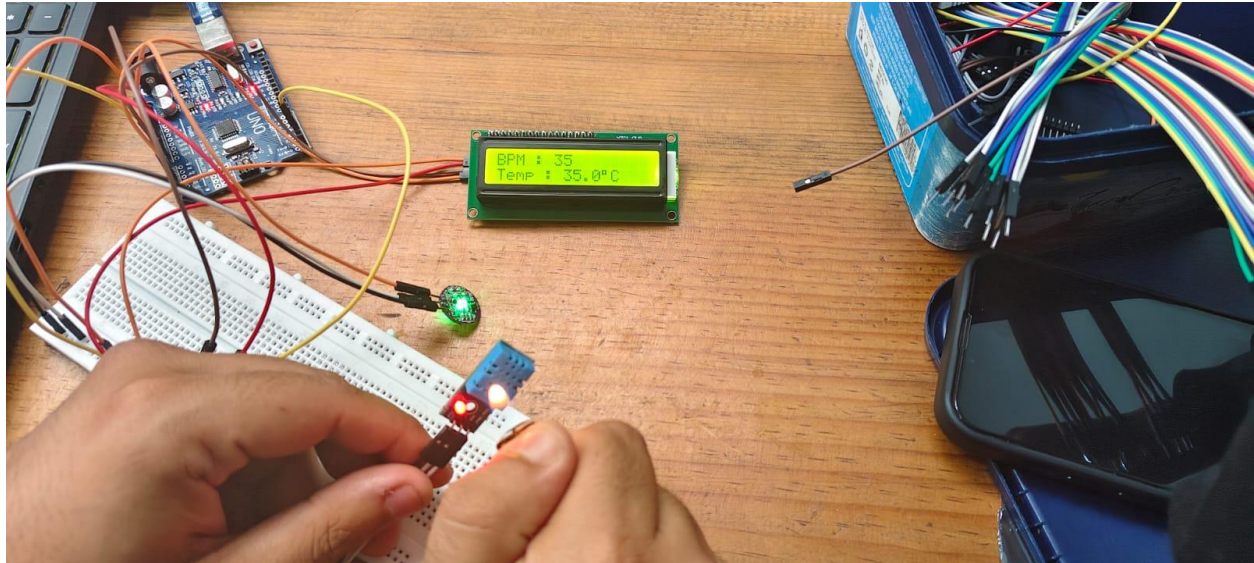


Figure 24: Temperature after over exposure of lighter

4.3.3 Testing 3: LCD Display testing

Table 7: Test 5 LCD display testing

Component	Test Objective	Expected Outcome	Actual outcome	Result
LCD display	To test if the LCD screen shows the string entered in the code.	The LCD should show the names of the group members as per the input kept in the code.	The LCD screen showed all the names of the members as coded.	Passed

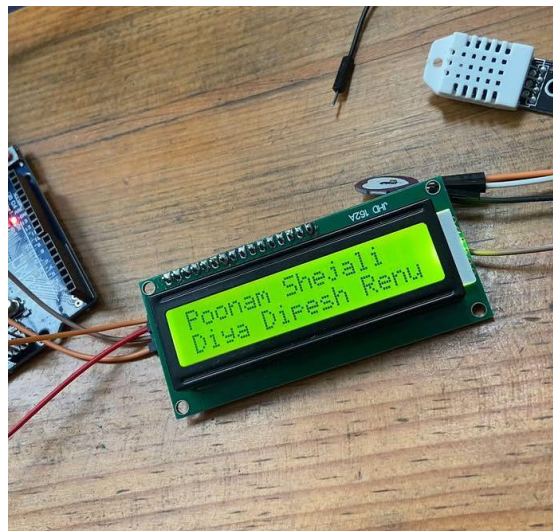
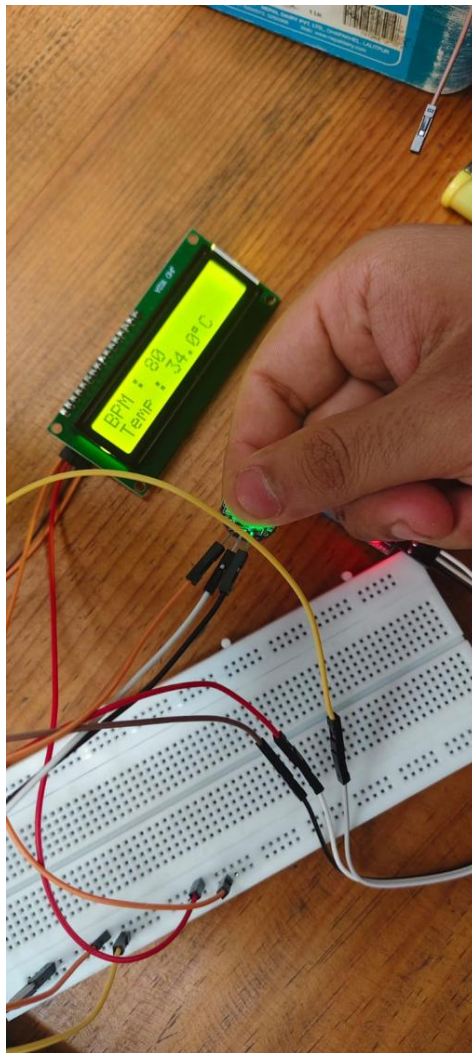


Figure 25: LCD display testing

4.3.4 Testing 4: Pulse Sensor testing

Component	Test Objective	Expected Outcome	Actual Outcome	Result
Pulse Sensor	To test if the pulse sensor reads the user's pulse properly.	The LCD screen should show the correct pulse readings of the user.	The LCD screen showed the correct pulse readings of the user.	Passed

*Figure 26: Pulse Sensor Testing*

4.4 Failed Test Cases

4.4.1 Pulse Sensor Failure Test Cases

Table 8 Pulse Sensor Failure Test Table

Failure scenario	System Response
Pulse sensor unplugged	The system responds with "Pulse Error".

4.4.1.1 Pulse Sensor Unplugged

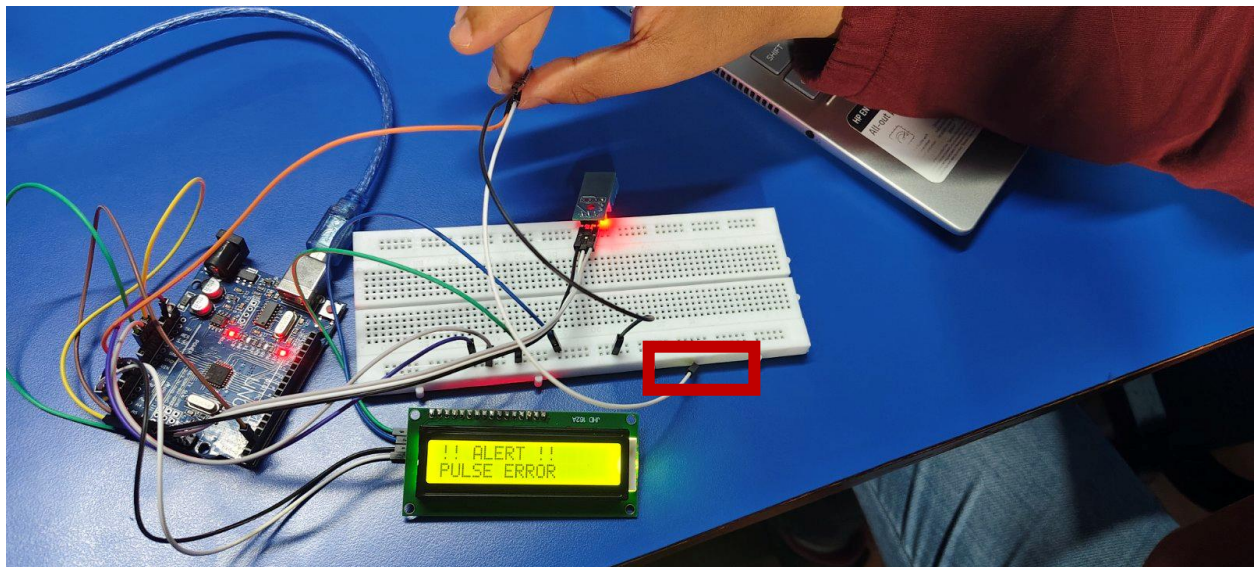


Figure 27: Pulse sensor VCC wire unplugged

4.4.2 DHT Sensor Failure Test Case

Table 9 DHT Sensor Failure Test Table

Failure scenario	System Response
DHT sensor unplugged	The system responds with "DHT Missing".

4.4.2.1 DHT Sensor Unplugged

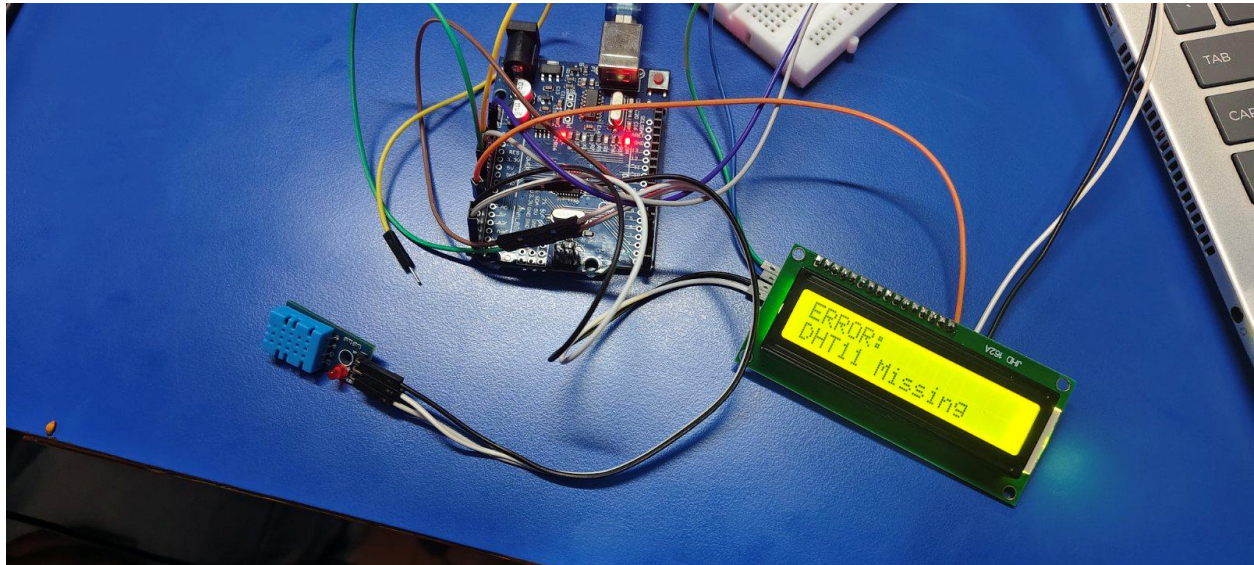


Figure 28: DHT11 VCC wire unplugged

5 Further Work

Although the developed system performs its intended functions effectively, there is still room for improvement in several areas. With additional time and resources, the system can be further enhanced to increase its accuracy, usability, and real-world applicability.

- Additional health sensors such as Electrocardiogram (ECG), Blood Oxygen Saturation (SPO2) and blood pressure monitors can be integrated to provide more detailed and reliable patient health data.
- The accuracy of existing sensors can be improved through proper calibration and by using higher-quality components.
- A dedicated mobile or web application can be developed to allow users and healthcare professionals to monitor data more easily in real time.
- Advanced technologies such as Artificial Intelligence (AI) and Machine Learning (ML) can be implemented to analyze data patterns and predict possible health risks.
- Data security can be strengthened by applying better encryption methods and secure consumption protocols to protect sensitive information.
- Power consumption can be optimized by using energy-efficient components and adding backup power solutions for uninterrupted operation.
- The system can be expanded to support multiple users, making it more suitable for hospitals and care centers.
- Cloud integration can be further improved for better data storage, management, and long-term analysis.
- Automated control features can be added, such as adjusting room temperature or ventilation based on sensor readings.
- The alert system can be enhanced by including multiple notification methods like SMS, email, or mobile alerts for quicker response.

(Chopade, 2023)

6 Conclusion

In conclusion, the health and room monitoring system aims to provide effective solutions for patients' appropriate care. By incorporating the use of different IOT components, this system can record the patient's health parameters and detect the presence of harmful gases to avoid any serious emergencies. Therefore, the modern world requires a system that benefits vulnerable patients.

This project shows the implementation and importance of sensors of pulse and temperature. It allows patients and healthcare professionals to react quickly to abnormal conditions, which in the end helps reduce potential health risks and improve patient safety.

Furthermore, the IoT for healthcare system promotes preventive healthcare by helping patients to identify early signs of any abnormalities or discomfort. With the data in backup and storing, it helps professionals to know about the abnormalities more quickly and reduce the health risk faster.

Overall, this project mostly highlights the improvement or potential of IoT in Healthcare system by making them more accurate, smarter, responsive and accessible. The developed IoT project can help with the problems, identify abnormalities faster, reduce possible health risk, provides individuals with health data and more.

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8 Appendices

8.1 Appendix A

8.2 Arduino UNO

It uses the ATmega328P (a tiny chip inside the Arduino UNO), microcontroller and comes with several digital and analog input/output pins that make it possible to connect different kinds of sensors and modules. It is the most popular favored microcontroller because of its simplicity and compatibility with various sensors (Arduino, 2023).

In this system, it collects data from the DHT11 sensor and pulse sensor, processes the readings, and displays the outcomes on the LCD.

8.3 Pulse Sensor

It operates according to the principles of photoplethysmography, which involves the use of light to detect the variations in blood volume beneath the skin layer. Once the user's finger is placed on the sensor, it captures the pulse signals and transmits them to the Arduino microcontroller. The Arduino then calculates the beats per minute (BPM) and displays the result on the LCD (Measure Heart Rate, 2026).

8.4 Liquid Crystal Display

The liquid crystals do not emit any light by themselves. Instead, they act as microscopic shutters that allow different amounts of light to pass through from the backlight. Depending upon the amount of electricity that is supplied to them, the liquid crystals align themselves, accordingly, letting different quantities of light pass through them. The light that is let through passes through three filters for creating colored images (Chris Woodford., 2024).

[Back to the top](#)

8.5 Appendix B

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
#include <DHT.h>
```

```
#include <PulseSensorPlayground.h>
```

8.6 DHT11 SETUP

```
#define DHTPIN 2
```

```
#define DHTTYPE DHT11
```

```
DHT dht(DHTPIN, DHTTYPE);
```

```
// =====
```

8.7 PULSE SENSOR SETUP

```
// =====
```

```
#define PULSE_PIN A1
```

```
#define THRESHOLD 550
```

```
PulseSensorPlayground pulseSensor;
```

8.8 LCD SETUP

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

8.9 VARIABLES

```
int myBPM = 0;
```

```
float temperature = 0.0;
```

```
float humidity = 0.0;
```

```
bool dhtError = false;
```

```
bool pulseError = false;
```

```
unsigned long lastDHTRead = 0;
```

```
unsigned long lastLCDUpdate = 0;
```

```
unsigned long lastBeatTime = 0;
```

```
bool showTempScreen = true;
```

8.10 SETUP

```
void setup() {
```

```
    Serial.begin(9600);
```

```
lcd.init();

lcd.backlight();


lcd.setCursor(0, 0);
lcd.print("Health Monitor");
lcd.setCursor(0, 1);
lcd.print("Starting...");
delay(2000);
lcd.clear();


dht.begin();


pulseSensor.analogInput(PULSE_PIN);
pulseSensor.setThreshold(THRESHOLD);


pulseSensor.begin();


Serial.println("System Ready!");
}
```

8.11 LOOP

```
void loop() {  
  
    // 1. PULSE SENSOR READ + ERROR CHECK  
  
    int pulseValue = analogRead(PULSE_PIN);  
  
    // Detect unplugged / floating pin  
    if (pulseValue < 10 || pulseValue > 1015) {  
        pulseError = true;  
        myBPM = 0;  
    } else {  
        pulseError = false;  
  
        if (pulseSensor.sawStartOfBeat()) {  
  
            int newBPM = pulseSensor.getBeatsPerMinute();  
  
            if (newBPM > 20 && newBPM < 220) {  
                myBPM = newBPM;  
                lastBeatTime = millis();  
            }  
        }  
    }  
}
```



```
}  
}  
  
// extra safety: no beat for long time = error  
if (millis() - lastBeatTime > 5000) {  
    pulseError = true;  
    myBPM = 0;  
}  
  
// 2. DHT11 READ EVERY 2 SECONDS  
  
if (millis() - lastDHTRead >= 2000) {  
  
    lastDHTRead = millis();  
  
    float newTemp = dht.readTemperature();  
    float newHum = dht.readHumidity();  
  
    if (isnan(newTemp) || isnan(newHum)) {  
  
        dhtError = true;  
  
        Serial.println("ERROR: DHT11 Unplugged!");  
    }  
}
```

```
} else {  
  
    dhtError = false;  
  
    temperature = newTemp;  
    humidity    = newHum;  
}  
  
}  
  
  
// 3. LCD UPDATE EVERY 1 SECOND  
  
if (millis() - lastLCDUpdate >= 1000) {  
  
    lastLCDUpdate = millis();  
  
    lcd.clear();  
  
  
    // PRIORITY 1: ERRORS
```

```
if (dhtError) {  
  
    lcd.setCursor(0, 0);  
    lcd.print("ERROR:");  
    lcd.setCursor(0, 1);  
    lcd.print("DHT11 Missing");  
    return;  
}
```

```
if (pulseError) {  
  
    lcd.setCursor(0, 0);  
    lcd.print("ERROR:");  
    lcd.setCursor(0, 1);  
    lcd.print("Pulse Missing");  
    return;  
}
```

```
// NORMAL DISPLAY
```

```
showTempScreen = !showTempScreen;
```

```
if (showTempScreen) {
```

```
    lcd.setCursor(0, 0);
```

```
    lcd.print("BPM: ");
```

```
    lcd.print(myBPM);
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print("Temp: ");
```

```
    lcd.print(temperature, 1);
```

```
    lcd.print((char)223);
```

```
    lcd.print("C");
```

```
}
```

```
else {
```

```
    lcd.setCursor(0, 0);
```

```
    lcd.print("BPM: ");
```

```
    lcd.print(myBPM);
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print("Hum: ");
```

```
    lcd.print(humidity, 1);  
    lcd.print("%");  
}  
  
delay(20);  
}
```

[Back to the top](#)